

EMG4_justice_trait_data

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packages

Questionnaire 2 data

```
## New names:
## * `1` -> `1...19`
## * `2` -> `2...20`
## * `1` -> `1...21`
## * `2` -> `2...22`

## # A tibble: 6 x 18
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1  8811     1     2     1     1     4     2     4     4     3     3     2     1
## 2  7735     1     4     1     1     1     1     1     3     2     1     2     1
## 3  7417     1     2     2     2     3     3     3     3     3     3     3     2
## 4  9080     1     3     1     1     2     1     4     4     4     1     2     1
## 5  6012     1     4     4     2     2     1     3     3     3     1     2     5
## 6  1145     2     4     1     1     1     1     1     2     2     4     4     3
## # i 5 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>
```

Select the questions that together measure the construct of Justice

```
justiceQs <- questionnaire_2[, -1]

# Cronbach's alpha justice questions
# Question 2, 12 and 13 were negatively correlated due to reverse framing. These questions w
ere automatically reversed.
reliability <- psych::alpha(justiceQs, check.keys=TRUE)
reliability

##
## Reliability analysis
## Call: psych::alpha(x = justiceQs, check.keys = TRUE)
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean   sd median_r
##   0.79      0.8      0.87      0.19 3.9 0.038  2.4 0.48      0.18
##
##   95% confidence boundaries
##           lower alpha upper
## Feldt      0.71  0.79  0.86
## Duhachek    0.72  0.79  0.87
##
## Reliability if an item is dropped:
##   raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## 1      0.79      0.79      0.87      0.19 3.9   0.038 0.029 0.19
```

```
## 2-      0.78      0.78      0.86      0.18 3.6      0.041 0.028 0.17
## 3      0.79      0.79      0.86      0.19 3.7      0.040 0.027 0.19
## 4      0.78      0.78      0.86      0.19 3.6      0.039 0.028 0.17
## 5      0.77      0.77      0.86      0.17 3.4      0.043 0.027 0.17
## 6      0.78      0.79      0.86      0.19 3.7      0.040 0.029 0.18
## 7      0.76      0.77      0.86      0.17 3.4      0.044 0.027 0.16
## 8      0.79      0.79      0.86      0.19 3.7      0.039 0.027 0.19
## 9      0.78      0.78      0.86      0.18 3.6      0.040 0.027 0.18
## 10     0.79      0.79      0.87      0.19 3.8      0.038 0.029 0.19
## 11     0.80      0.80      0.87      0.20 4.0      0.037 0.025 0.19
## 12-     0.79      0.79      0.86      0.19 3.7      0.039 0.025 0.19
## 13-     0.79      0.80      0.86      0.20 3.9      0.038 0.024 0.20
## 14     0.80      0.80      0.87      0.20 3.9      0.037 0.028 0.19
## 15     0.77      0.78      0.85      0.18 3.5      0.042 0.026 0.17
## 16     0.78      0.78      0.85      0.18 3.6      0.041 0.026 0.18
## 17     0.78      0.78      0.86      0.18 3.5      0.040 0.028 0.17
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## 1    60 0.28 0.37 0.30 0.24 1.2 0.38
## 2-   60 0.57 0.55 0.51 0.47 3.5 1.03
## 3    60 0.45 0.47 0.44 0.36 1.7 0.82
## 4    60 0.47 0.50 0.47 0.38 1.8 0.89
## 5    60 0.68 0.67 0.65 0.60 2.6 1.09
## 6    60 0.47 0.49 0.44 0.39 1.6 0.81
## 7    60 0.71 0.67 0.65 0.62 2.6 1.18
## 8    60 0.46 0.46 0.43 0.34 2.8 1.12
## 9    60 0.53 0.50 0.48 0.44 2.5 0.93
## 10   60 0.38 0.40 0.34 0.28 1.7 0.93
## 11   60 0.27 0.29 0.24 0.15 2.5 0.98
## 12-  60 0.45 0.44 0.42 0.34 4.3 1.03
## 13-  60 0.35 0.34 0.32 0.23 4.0 1.04
## 14   60 0.33 0.34 0.27 0.20 2.2 1.14
## 15   60 0.64 0.61 0.61 0.54 2.3 1.10
## 16   60 0.59 0.55 0.54 0.47 2.5 1.31
## 17   60 0.56 0.59 0.56 0.49 1.6 0.74
##
## Non missing response frequency for each item
##      1      2      3      4      5 miss
## 1  0.83 0.17 0.00 0.00 0.00 0
## 2  0.10 0.53 0.13 0.20 0.03 0
## 3  0.50 0.38 0.07 0.05 0.00 0
## 4  0.45 0.42 0.05 0.08 0.00 0
## 5  0.15 0.40 0.23 0.17 0.05 0
## 6  0.57 0.33 0.05 0.05 0.00 0
## 7  0.12 0.48 0.12 0.20 0.08 0
## 8  0.12 0.32 0.28 0.22 0.07 0
## 9  0.08 0.48 0.25 0.17 0.02 0
## 10 0.50 0.38 0.07 0.02 0.03 0
## 11 0.15 0.42 0.27 0.15 0.02 0
## 12 0.58 0.22 0.12 0.07 0.02 0
## 13 0.38 0.37 0.13 0.10 0.02 0
## 14 0.33 0.35 0.15 0.13 0.03 0
## 15 0.20 0.52 0.10 0.13 0.05 0
## 16 0.23 0.43 0.05 0.18 0.10 0
## 17 0.50 0.38 0.10 0.02 0.00 0
```

The overall Cronbach's alpha is 0.7940737, so we include all questions in our composite Justice score. We do need to reverse the scores for questions 2, 12 and 13.

```
## # A tibble: 6 x 18
##   sjCode `1` `2` `3` `4` `5` `6` `7` `8` `9` `10` `11` `12`
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1  8811      1      2      1      1      4      2      4      4      3      3      2      1
## 2  7735      1      4      1      1      1      1      1      3      2      1      2      1
## 3  7417      1      2      2      2      3      3      3      3      3      3      3      2
```

```
## 4 9080 1 3 1 1 2 1 4 4 4 1 2 1
## 5 6012 1 4 4 2 2 1 3 3 3 1 2 5
## 6 1145 2 4 1 1 1 1 1 1 2 2 4 3
## # i 5 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## # `17` <dbl>
```

dataframe with only mean justice rating scores (trait score) per participant

```
## # A tibble: 6 x 2
##   sjCode J_score
##   <dbl>   <dbl>
## 1 1055 0.235
## 2 1088 0.882
## 3 1124 1.06
## 4 1145 1.18
## 5 1258 1
## 6 1278 1.18

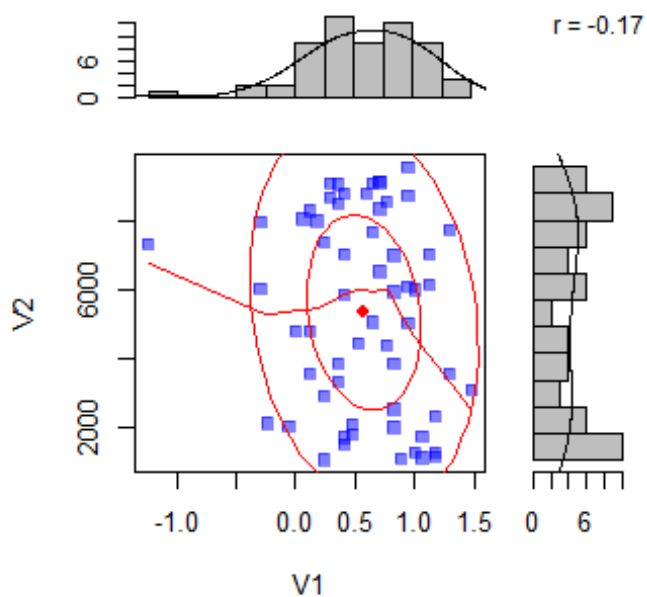
summary(questionnaire_2_means$J_score)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1.2353  0.2941   0.6471  0.5716  0.9412  1.4706

sd(questionnaire_2_means$J_score)

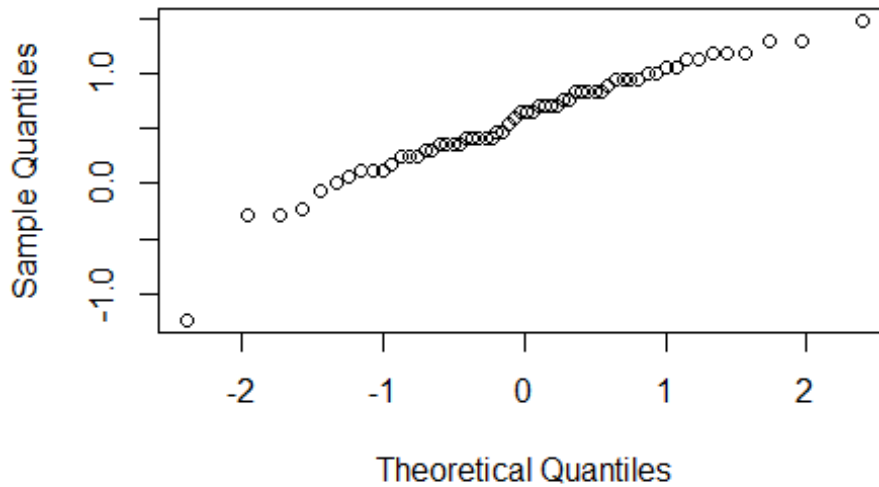
## [1] 0.4787541

scatterHist(questionnaire_2_means$J_score,questionnaire_2_means$sjCode)
```



```
# https://www.reneshbedre.com/blog/find-outliers.html using Median Absolute Deviation (MAD)
qqnorm(questionnaire_2_means$J_score)
```

Normal Q-Q Plot



```
# get median
med = median(questionnaire_2_means$J_score)
# subtract median from each value of x and get absolute deviation
abs_dev = abs(questionnaire_2_means$J_score-med)
# get MAD
mad = 1.4826 * median(abs_dev)

# get threshold values for outliers
Tmin = med-(3*mad)
Tmax = med+(3*mad)

# find outlier
outlier <- questionnaire_2_means$J_score[which(questionnaire_2_means$J_score < Tmin | questionnaire_2_means$J_score > Tmax)]
```

The Justice score -1.2352941 is an outlier, therefore, we plot another histogram without the outlier.

```
questionnaire_2_means_no_outlier <- questionnaire_2_means %>%
  filter(J_score > -1)

scatterHist(questionnaire_2_means_no_outlier$J_score,questionnaire_2_means_no_outlier$sjCode
)
```

